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Miniature telephoto lens module and a camera utilizing such a lens module

Abstract

The presently disclosed subject matter includes a mobile electronic comprising an integrated camera, comprising a Wide camera unit comprising a Wide lens unit, and a Telephoto camera unit comprising a telephoto lens unit, the telephoto lens unit and the wide lens unit having respectively TTL/EFL ratios smaller and larger than 1 and defining separate telephoto and wide optical paths.

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2010/0196001	12/2009	Ryynänen et al.	N/A	N/A
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2010/0238327	12/2009	Griffith et al.	N/A	N/A
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2011/0149119	12/2010	Matsui	N/A	N/A
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2011/0164172	12/2010	Shintani et al.	N/A	N/A
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2012/0069235	12/2011	Imai	N/A	N/A
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2013/0002928	12/2012	Imai	N/A	N/A
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2015/0323757	12/2014	Bone	N/A	N/A
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2016/0154066	12/2015	Hioka et al.	N/A	N/A
2016/0154202	12/2015	Wippermann et al.	N/A	N/A
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Background/Summary

CROSS REFERENCE TO EXISTING APPLICATIONS (1) This application is a Continuation application from U.S. patent application Ser. No. 18/641,478 filed Apr. 22, 2024 (now allowed), which was a Continuation application from U.S. patent application Ser. No. 17/389,510 filed Jul. 30, 2021 (now U.S. Pat. No. 11,994,654), which was a Continuation application from U.S. patent application Ser. No. 16/276,034 filed Feb. 14, 2019 (now U.S. Pat. No. 11,125,975), which was a Continuation application from U.S. patent application Ser. No. 15/540,676 filed Jun. 29, 2017 (now U.S. Pat. No. 10,288,840), which was a 371 from international patent application No. PCT/IB2015/050044 filed Jan. 3, 2015.

TECHNOLOGICAL FIELD

(1) The present invention is generally in the field of imaging techniques, and relates to a camera and mobile electronic devices utilizing such a camera.

BACKGROUND

(2) Digital camera modules are currently being incorporated into a variety of portable electronic devices. Such devices include for example mobile phones (e.g. smartphones), personal data assistants (PDAs), computers, and so forth. Digital camera modules for use in portable devices have to meet certain requirements such as good quality imaging, small footprint, as well as low weight.

(3) Several techniques for small digital camera modules providing good quality imaging are described in WO14083489 and WO14199338, both assigned to the assignee of the present application.

(4) According to the technique described in WO14083489, a multi-aperture imaging system comprises a first camera with a first sensor that captures a first image and a second camera with a second sensor that captures a second image. The two cameras have either identical or different FOVs. Either image may be chosen to be a primary or an auxiliary image, based on a zoom factor. An output image with a point of view determined by the primary image is obtained by registering the auxiliary image to the primary image.

(5) The technique described in WO14199338 relates to a dual-aperture zoom digital camera operable in both still and video modes. The camera includes Wide and Tele imaging sections with respective lens/sensor combinations and image signal processors and a camera controller operatively coupled to the Wide and Tele imaging sections. The controller is configured to combine in still mode at least some of the Wide and Tele image data to provide a fused output image from a particular point of view, and to provide, without fusion, continuous zoom video mode output images, each output image having a given output resolution. The video mode output images are provided with a smooth transition when switching between a lower zoom factor (ZF) value and a

higher ZF value or vice versa. At the lower ZF the output resolution is determined by the Wide sensor, while at the higher ZF value the output resolution is determined by the Tele sensor.

GENERAL DESCRIPTION

(6) There is a need in the art for a novel camera module for use in modern portable electronic devices, such as smart phones, laptops, notepads, etc.

(7) As noted above, the requirements for the camera modules for use in such devices are related to the size, weight and image quality of the camera. Moreover, these requirements become more essential when the camera module is to be installed within the portable device, unlike other external camera units attachable to the portable device. In the case of an internal (integral) camera unit, the dimensions of the camera optics should be as small as possible in order to be suitable to operate with commonly used detectors and to fit the thickness of the device in which the camera is installed (preferably without protruding from the device's casing), while the trend in such devices is to reduce the thickness as much as possible.

(8) This problem is even more crucial when using, in a portable device, a lens with a long length with a fixed and relatively high zooming effect. Considering for example the dual-aperture zoom digital camera described in above-indicated publications WO14083489 and WO14199338 mentioned above, it utilizes Wide and Tele imaging channels which provide advanced imaging capabilities such as zoom and image quality by image fusion between the two channels.

(9) One of the problems with dual-aperture zoom cameras relates to the dimensions (heights) of Wide and Tele cameras along the optical axis. Such dimensions depend on total track lengths (TTLs) of the Tele and Wide lenses used in the respective imaging channels.

(10) As schematically illustrated in FIG. 1B, the TTL is typically defined as the maximal distance between the object-side surface of the lens module and an image plane IP defined by such a lens module (where the sensing surface of a camera detector is placed). In most miniature lenses, the TTL is larger than the effective focal length (EFL) of the lens module, which is equal to the distance between the effective principal plane of the lens and its focal plane (which substantially coincides with image plane IP).

(11) With regard to the term effective principal plane, the following should be understood. Generally, the lens (or lens module) has front and rear principal planes, which have the property that a ray emerging from the lens appears to have crossed the rear principal plane at the same distance from the axis that that ray appeared to cross the front principal plane, as viewed from the front of the lens. This means that the lens can be treated as if all of the refraction occurred at the principal planes. The principal planes are crucial in defining the optical properties of the system, since it is the distance of the object and image from the front and rear principal planes that determine the magnification of the system. The principal points are the points where the principal planes cross the optical axis.

(12) Considering dual-aperture optical zoom in a mobile phone (e.g. a smartphone) with the typically used lenses, i.e. typical TTL/EFL ratio of about 1.3, the Wide and Tele lenses would have TTLs of about 4.55 mm and 9.1 mm, respectively. This will result in undesirably long camera modules for use in such a smartphone device.

(13) Further, the difference in the TTLs of the Wide and Tele lens modules can cause shadowing and light-blocking problems. Reference is made to FIG. 1A schematically illustrating that part of incoming light incident on the “higher” lens does not reach the “shorter” lens. In this connection, one should keep in mind that a distance between the Tele and Wide lens modules should be as small as possible to meet the overlapping/common FOVs as well as footprint requirements for the camera unit in a portable device.

(14) Another part of the presently disclosed subject matter is associated with the implementation of standard optical image stabilization (OIS) in a dual-aperture zoom camera. Standard OIS compensates for camera tilt (“CT”) by a parallel-to-the image sensor (exemplarily in the X-Y plane) lens movement (“LMV”). Camera tilt causes image blur. The amount of LMV (in mm)

needed to counter a given camera tilt depends on the cameras lens EFL, according to the relation $LMV=CT \cdot EFL$ where "CT" is in radians and EFL is in mm. Since, as shown above, a dual-aperture zoom camera may include two lenses with significantly different EFLs, it is impossible to move both lenses together and achieve optimal tilt compensation for both Tele and Wide cameras. That is, since the tilt is the same for both cameras, a movement that will cancel the tilt for the Wide camera will be insufficient to cancel the tilt for the Tele camera. Similarly, a movement that will cancel the tilt for the Tele camera will over-compensate the tilt cancellation for the Wide camera. Assigning a separate OIS actuator to each camera can achieve simultaneous tilt compensation, but at the expense of a complicated and costly camera system.

(15) Thus, for both a single-aperture or multi-aperture (dual) camera unit, the use of a telephoto lens would be advantageous, as such a telephoto lens provides reduced TTL while enabling to maintain the relatively high EFL required for the Tele lens, i.e. for telephoto lens $TTL < EFL$. However, the dimensions of conventional lenses in which the telephoto condition is satisfied do not allow them to be used as integral lenses fully embedded in a thin portable device. The telephoto lens module, in order to be used as an integral lens in a modern portable device, has to satisfy the telephoto condition (i.e. $TTL < EFL$) while the lens module is to be as short as possible (along the optical path of light passing through it) allowing it to be fully fitted within the portable device casing.

(16) Accordingly, a miniature telephoto lens module is disclosed which is designed with the desired dimensions to enable its integration within a portable device. According to some examples of the presently disclosed subject matter, the miniature telephoto lens module (or telephoto lens unit) is designed to be completely integrated within the casing of a conventional Smartphone, i.e. without protruding therefrom. The disclosed telephoto lens module has a total track lens (TTL) smaller than an effective focal lens (EFL) thereof, and is configured such that its dimension along the optical axis is desirably small, i.e. about 4-15 mm or less (e.g. suitable to be fitted in a portable device having a casing as small as 4 mm).

(17) The telephoto lens unit comprises multiple lens elements made of at least two different polymer materials having different Abbe numbers. The multiple lens elements comprise a first group of at least three lens elements being a telephoto lens assembly, and a second group of at least two lens elements being a field lens assembly.

(18) The first group of lens elements comprises, in order from the object plane to the image plane along an optical axis of the telephoto lens unit: a first lens having positive optical power and a pair of second and third lenses having together negative optical power such that said telephoto lens assembly provides a telephoto optical effect of said telephoto lens unit and wherein said second and third lenses are each made of one of said at least two different polymer materials having a different Abbe number, for reducing chromatic aberrations of said telephoto lens. The second group of lens elements is configured to correct field curvature of said telephoto lens assembly, and said field lens module comprises two or more of said lens elements made of the different polymer materials respectively having different Abbe numbers, and configured to compensate for residual chromatic aberrations of said telephoto lens assembly dispersed during light passage through an effective gap located between the telephoto and field lens assemblies. The effective gap is larger than $\frac{1}{5}$ of the TTL of the telephoto lens unit, thereby allowing sufficient field separation for reducing chromatic aberration.

(19) Various examples disclosed herein include an optical lens unit comprising, in order from an object side to an image side: a first lens element with positive refractive power having a convex object-side surface, a second lens element with negative refractive power having a thickness $d_{sub.2}$ on an optical axis and separated from the first lens element by a first air gap, a third lens element with negative refractive power and separated from the second lens element by a second air gap, a fourth lens element having a positive refractive power and separated from the third lens element by an effective third air gap, and a fifth lens element having negative refractive power,

separated from the fourth lens element by an effective fourth air gap, the fifth lens element having a thickness $d_{sub.5}$ on the optical axis.

(20) An optical lens unit may further include a stop, positioned before the first lens element, a glass window disposed between the image-side surface of the fifth lens element and an image sensor with an image plane on which an image of the object is formed.

(21) Each lens element has two surfaces, each surface having a respective diameter. The largest diameter among all lens elements is defined as an “optical diameter” of the lens assembly.

(22) As disclosed herein, TTL is defined as the distance on an optical axis between the object-side surface of the first lens element and an image plane where the image sensor is placed. “EFL” has its regular meaning, as mentioned above. In all embodiments, TTL is smaller than the EFL, i.e. the TTL/EFL ratio is smaller than 1.0. In some embodiments, the TTL/EFL ratio is smaller than 0.9. In an embodiment, the TTL/EFL ratio is about 0.85. According to some examples the lens assembly has an F number $F \# < 3.2$.

(23) According to an example disclosed herein, the focal length of the first lens element f_1 is smaller than $TTL/2$, the first, third and fifth lens elements have each an Abbe number (“Vd”) greater than 50, the second and fourth lens elements have each an Abbe number smaller than 30, the first air gap is smaller than $d_{sub.2}/2$, the effective third air gap is greater than $TTL/5$ and the effective fourth air gap is smaller than $1.5d_{sub.5}TTL/50$. In some embodiments, the surfaces of the lens elements may be aspheric.

(24) In the optical lens unit mentioned above, the first lens element with positive refractive power allows the TTL of the lens unit to be favorably reduced. The combined design of the first, second and third lens elements plus the relative short distances between them enable a long EFL and a short TTL. The same combination, together with the high dispersion (low Vd) for the second lens element and low dispersion (high Vd) for the first and third lens elements, also helps to reduce chromatic aberration. In particular, the ratio $TTL/EFL < 1.0$ and minimal chromatic aberration are obtained by fulfilling the relationship $1.2 \times |f_3| > |f_2| > 1.5 \times f_1$, where “f” indicates the lens element effective focal length and the numerals 1, 2, 3, 4, 5 indicate the lens element number.

(25) The relatively large effective gap between the third and the fourth lens elements plus the combined design of the fourth and fifth lens elements assist in bringing all fields' focal points to the image plane. Also, because the fourth and fifth lens elements have different dispersions and have respectively positive and negative power, they help in minimizing chromatic aberration.

(26) The telephoto lens module disclosed herein may be advantageously adapted to be incorporated in a mobile phone camera that uses a typical $\frac{1}{4}$ " or $\frac{1}{3}$ " image sensor. For example, to be competitive with known mobile phone cameras with $\frac{1}{4}$ " image sensors, it would be advantageous for the TTL of the telephoto lens module to be smaller than 5.5 mm and the largest lens diameter to be smaller than 4 mm. To be competitive with known mobile phone cameras with $\frac{1}{3}$ " image sensors, it would be advantageous for the TTL of the telephoto lens module to be smaller than 6.5 mm and the largest lens diameter to be smaller than 5 mm.

(27) Accordingly to an example of the presently disclosed subject matter there is provided an optical lens unit configured to provide an image on an entire area of a $\frac{1}{4}$ " image sensor, the lens unit comprising five lens elements and having a TTL smaller than 5.5 mm, an EFL larger than 5.9 mm, and an optical diameter smaller than 4 mm.

(28) Accordingly in another example of the presently disclosed subject matter there is provided an optical lens unit operative to provide an image on an entire area of a $\frac{1}{3}$ " image sensor, the lens unit comprising five lens elements and having a TTL smaller than 6.2 mm, an EFL larger than 6.8 mm, and an optical diameter smaller than 5 mm.

(29) Also, as mentioned above, according to the presently disclosed subject matter it is suggested to have all lens elements made of polymer material such as plastic. While lenses made of polymer material are advantageous for reducing the price tag of the telephoto lens module as well as its weight, there are very few polymer materials which are suitable for this purpose. This is different to

glass lenses which can be made of a variety of different glass materials, each characterized by a different Abbe number. The scarcity in polymer materials presents a challenge when designing lenses for a telephoto lens module. This challenge is at least partly due to the limitation in possible combinations of different lenses with different Abbe numbers which can be used for the purpose of correcting field curvature and compensating for chromatic aberrations.

(30) Thus, according to one aspect of the presently disclosed subject matter there is provided a mobile electronic device comprising an integrated camera, wherein the camera comprises a Wide camera unit comprising a Wide lens unit, and a Telephoto camera unit comprising a telephoto lens unit, the telephoto lens unit and the wide lens unit having respectively TTL/EFL ratios smaller and larger than 1 and defining separate telephoto and wide optical paths.

(31) In addition to the above features, the mobile electronic device according to this aspect of the presently disclosed subject matter can optionally comprise one or more of features (i) to (xvi) below, in any desired combination or permutation: (i). wherein light receiving outer surfaces of the Wide and Telephoto lens units are located substantially in the same plane, thereby reducing shadowing and light blocking effects therebetween. (ii). wherein the Wide and Telephoto camera units are mounted on separate printed circuit boards. (iii). wherein the printed circuit boards are located in different spaced-apart substantially parallel planes. (iv). wherein the Wide and Telephoto camera units are mounted directly on a single printed circuit board. (v). wherein the Wide and Telephoto camera units are spaced from one another a distance d of about 1 mm. (vi). wherein the telephoto lens unit is made of at least two polymer materials. (vii). wherein the telephoto lens has a total track lens (TTL) not exceeding 15 mm. (viii). wherein the telephoto lens has TTL less than 10 mm. (ix). wherein the telephoto lens unit comprises multiple lens elements made of at least two different polymer materials having different Abbe numbers, the multiple lens elements comprise a first group of at least three lens elements configured to form a telephoto lens assembly, and a second group of at least two lens elements configured to form a field lens assembly, wherein the field lens assembly is spaced from the telephoto lens assembly by a predetermined effective gap. (x). wherein said at least two different polymer materials comprise at least one plastic material with the Abbe number larger than 50, and at least one plastic material with the Abbe number smaller than 30. (xi). wherein the first group of lens elements comprises, in order from an object plane to an image plane along an optical axis of the telephoto lens unit: a first lens having positive optical power and a pair of second and third lenses having together negative optical power such that said telephoto lens assembly provides telephoto optical effect of said telephoto lens unit, and said second and third lenses are each made of one of said at least two different polymer materials having a different Abbe number, for reducing chromatic aberrations of said telephoto lens; and the second group of lens elements is configured to correct field curvature of said telephoto lens assembly, and comprises two or more of said lens elements made of the different polymer materials respectively having different Abbe numbers, and configured to compensate for residual chromatic aberrations of said telephoto lens assembly dispersed during light passage through said effective gap between the telephoto and field lens assemblies. (xii). wherein the first, third and fifth lens elements have each an Abbe number greater than 50, and the second and fourth lens elements have each an Abbe number smaller than 30. (xiii). wherein the predetermined effective gap is equal to or larger than $\frac{1}{5}$ of the TTL of the telephoto lens unit. (xiv). wherein the lens elements of the field lens assembly are spaced from one another an effective air gap smaller than $\frac{1}{50}$ of the TTL of the telephoto lens unit. (xv). wherein the telephoto lens unit has a TTL smaller than 5.5 mm, an effective focal length (EFL) larger than 5.9 mm, and an optical diameter smaller than 4 mm, thereby enabling to provide an image on an entire area of a $\frac{1}{4}$ " image sensor. (xvi). wherein the telephoto lens unit has a TTL smaller than 6.2 mm, an effective focal length (EFL) larger than 6.8 mm, and an optical diameter smaller than 5 mm, thereby enabling to provide an image on an entire area of a $\frac{1}{3}$ " image sensor.

(32) According to another aspect of the presently disclosed subject matter there is provided a

camera for integrating in a mobile electronic device, the camera comprising a Wide camera unit and a Telephoto camera unit comprising respectively a wide lens unit and a telephoto lens unit having TTL/EFL ratios larger and smaller than 1, respectively, and defining wide and telephoto optical paths.

(33) Wherein according to some examples the lens elements of at least the telephoto lens unit are made of one or more polymer materials.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to better understand the subject matter that is disclosed herein and to exemplify how it may be carried out in practice, embodiments will now be described, by way of non-limiting example only, with reference to the accompanying drawings, in which:

(2) FIG. 1A is a schematic illustration demonstrating shadowing and light-blocking problems caused by height differences between Wide and Tele cameras in a dual-aperture camera;

(3) FIG. 1B is a schematic illustration of a mobile phone device (constituting a portable electronic device) utilizing a camera unit as disclosed herein which is fully integrated inside the smartphone device;

(4) FIG. 1C is a schematic illustration of a telephoto lens unit according to the presently disclosed subject matter;

(5) FIG. 2A is a schematic illustration of a specific configuration of the telephoto lens unit, according to a first example of the presently disclosed subject matter;

(6) FIG. 2B shows a graph plotting the modulus of the optical transfer function (MTF) vs. focus shift of the entire optical lens unit of FIG. 2A for various fields;

(7) FIG. 2C shows a graph plotting the distortion vs. field angle (+Y direction) for the lens unit of FIG. 2A;

(8) FIG. 3A is a schematic illustration of another possible configuration of the telephoto lens unit, according to a first example of the presently disclosed subject matter;

(9) FIG. 3B shows a graph plotting the MTF vs. focus shift of the entire optical lens assembly for various fields in the lens unit of FIG. 3B, according to the second example of the presently disclosed subject matter;

(10) FIG. 3C shows a graph plotting the distortion +Y in percent for the lens unit of FIG. 3A;

(11) FIG. 4A is a schematic illustration of a specific configuration of the telephoto lens unit, according to a first example of the presently disclosed subject matter;

(12) FIG. 4B shows a graph plotting the MTF vs. focus shift of the entire optical lens system for various fields in the lens unit of FIG. 4A;

(13) FIG. 4C shows a graph plotting the distortion +Y in percent for the lens unit of FIG. 4A;

(14) FIG. 5 is a schematic illustration showing the concept of an effective air gap between adjacent lenses in an optical lens unit, according to the presently disclosed subject matter;

(15) FIG. 6A is a schematic illustration, in perspective cross section, of an example of a dual-aperture zoom camera, with each camera on a separate printed circuit board (PCB), according to the presently disclosed subject matter;

(16) FIG. 6B is a schematic illustration, in perspective cross section, of another example of a dual-aperture zoom camera, with each camera on a separate PCB, according to the presently disclosed subject matter;

(17) FIG. 7 is a schematic illustration, in perspective cross section, of yet another example of a dual-aperture zoom camera, where both cameras are mounted on a single PCB, according to the presently disclosed subject matter;

(18) FIG. 8 is a schematic illustration of an example of a dual-aperture zoom camera that includes

an OIS mechanism, according to the presently disclosed subject matter; and

(19) FIG. 9 shows schematically a functional block diagram of the camera example of FIG. 8, according to the presently disclosed subject matter.

DETAILED DESCRIPTION OF EMBODIMENTS

(20) The present invention includes novel configuration of a lens unit in a portable camera, advantageously applicable in a portable electronic device. This is schematically illustrated in FIG. 1B. In this example, such a portable electronic device **10** is constituted by a mobile phone device (e.g. smartphone). The mobile device is typically a few millimeters thick, e.g. 4 mm-15 mm.

(21) However, as explained above and exemplified further below, the problems solved by the technique disclosed herein are relevant for any modern electronic device equipped with a camera **15** and suitable to be implemented in any such device. This is so since any modern electronic device of the kind specified (i.e. a device including an integral camera unit) is to be as slim as possible, as light as possible, and is to acquire pictures with as good quality as possible.

(22) Modern cameras typically require zooming functions. When such a camera is used in an electronic device, such as a mobile phone device, the zooming function is often implemented with static optics. The problems which may arise when trying to incorporate Wide and Tele lenses into a common housing due to the difference in their heights are described above with reference to FIG. 1A.

(23) As mentioned above, the presently disclosed subject matter includes a novel mobile electronic device **10** which includes an integrated camera unit **15** which is mounted inside the device casing **14**. The camera **15** includes at least one telephoto lens unit (not shown here) which is made of polymer materials. The telephoto lens unit is configured such that its total track lens (TTL) is less than 15 mm and even less than 10 mm, e.g. less than 6 mm or even less than 4 mm. Thus, enabling the camera to be fully integrated in the portable device (substantially not protruding from the device casing).

(24) Reference is made to FIG. 1C showing schematically the configuration of a telephoto lens unit **20** of the present invention. The telephoto lens unit **20** is composed of multiple lens elements made of different polymer materials, i.e. materials having different Abbe numbers. The multiple lens elements are configured and arranged to define a telephoto lens assembly **22A** and a field lens assembly **22B** arranged along an optical axis OA with a predetermined effective gap G between them (as will be described more specifically further below). The telephoto lens assembly **22A** is configured to provide the telephoto optical effect of the telephoto lens unit **20**. The field lens assembly **22B** spaced from the telephoto lens assembly **22A** by the predetermined effective gap G is configured for correcting field curvature of the telephoto lens assembly **22A** and to compensate for residual chromatic aberrations of the telephoto lens assembly dispersed during light passage through the effective gap G.

(25) The telephoto lens unit **20** is characterized by a total track lens (TTL) and an effective focal lens (EFL) such that $TTL < EFL$. This will be exemplified further below. According to the invention, the effective gap G between assemblies **22A** and **22B** is selected to be larger than $TTL/5$ of the telephoto lens unit **22A**, thereby enabling correction of field curvature of telephoto lens assembly **22A** by the field lens assembly **22B**.

(26) The telephoto lens assembly **22A** includes three lens elements (generally three or more) L1, L2, L3 (which are shown here schematically and not to scale), where lens L1 has positive optical power and lenses L2 and L3 have together negative optical power. Lenses L2 and L3 are made of the first polymer material having a first Abbe number selected for reducing chromatic aberrations of the telephoto lens assembly **22A**. The field lens assembly **22B** includes two (or more) lens elements L4 and L5 which are made of different polymer materials respectively having different Abbe numbers. These lenses are configured to compensate for residual chromatic aberrations of the telephoto lens assembly **22A** dispersed during light passage through the effective gap G between the **22A** and **22B**.

(27) Lenses L1-L5 can be made for example of two plastic materials, one having an Abbe number greater than 50 and the other smaller than 30. For example, Lenses L1, L3 and L5 are made of plastic with an Abbe number greater than 50, and lenses L2 and L4 are made of plastic having an Abbe number smaller than 30.

(28) The following are several specific, but non-limiting, examples of the implementation and operation of the telephoto lens unit of the invention described above with reference to FIG. 1C. In the following description, the shape (convex or concave) of a lens element surface is defined as viewed from the respective side (i.e. from an object side or from an image side).

(29) FIG. 2A shows a schematic illustration of an optical lens unit 100, according to a first example of the presently disclosed subject matter. FIG. 2B shows the MTF vs. focus shift of the entire optical lens unit for various fields in the lens unit configuration 100. FIG. 2C shows the distortion +Y in percent vs. field.

(30) According to the example illustrated in FIG. 2A, lens unit 100 includes, in order from an object side to an image side, a first plastic lens element 102 (also referred to as “L1”) with positive refractive power having a convex object-side surface 102a and a convex or concave image-side surface 102b; a second plastic lens element 104 (also referred to as “L2”) with negative refractive power and having a meniscus convex object-side surface 104a, with an image side surface marked 104b; a third plastic lens element 106 (also referred to as “L3”) with negative refractive power having a concave object-side surface 106a with an inflection point and a concave image-side surface 106b. These lens elements define together the telephoto lens assembly (22A in FIG. 1C). Further provided in lens unit 100 is a fourth plastic lens element 108 (also referred to as “L4”) with positive refractive power having a positive meniscus, with a concave object-side surface marked 108a and an image-side surface marked 108b; and a fifth plastic lens element 110 (also referred to as “L5”) with negative refractive power having a negative meniscus, with a concave object-side surface marked 110a and an image-side surface marked 110b. These two lenses define together the field lens assembly (22B in FIG. 1C). The optical lens unit 100 may further optionally include a stop element 101. The telephoto lens unit 100 defines an image plane 114 in which image sensor(s) is/are located, which is not shown here. Also, as exemplified in the figure, an optional glass window 112 is disposed between the image-side surface 110b of fifth lens element 110 and the image plane 114.

(31) In the example of the telephoto lens unit 100, all lens element surfaces are aspheric. Detailed optical data is shown in Table 1, and aspheric surface data is shown in Table 2, wherein the units of the radius of curvature (R), lens element thickness and/or distances between elements along the optical axis and diameter are expressed in mm. “Nd” is the refraction index. The equation of the aspheric surface profiles is expressed by:

(32)
$$z = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2r^2}} + \alpha_1 r^2 + \alpha_2 r^4 + \alpha_3 r^6 + \alpha_4 r^8 + \alpha_5 r^{10} + \alpha_6 r^{12} + \alpha_7 r^{14}$$
 where r is the distance from (and is perpendicular to) the optical axis, k is the conic coefficient, c=1/R where R is the radius of curvature, and α are coefficients given in Table 2.

(33) In the equation above as applied to the telephoto lens unit, coefficients $\alpha_{\text{sub.1}}$ and $\alpha_{\text{sub.7}}$ are zero. It should be noted that the maximum value of r “max r”=Diameter/2. It should also be noted that in Table 1 (and in Tables 3 and 5 below), the distances between various elements (and/or surfaces) are marked “Lmn” (where m refers to the lens element number, n=1 refers to the element thickness and n=2 refers to the air gap to the next element) and are measured on the optical axis z, wherein the stop is at z=0. Each number is measured from the previous surface. Thus, the first distance -0.466 mm is measured from the stop to surface 102a, the distance L11 from surface 102a to surface 102b (i.e. the thickness of first lens element 102) is 0.894 mm, the air gap L12 between surfaces 102b and 104a is 0.020 mm, the distance L21 between surfaces 104a and 104b (i.e. thickness d2 of second lens element 104) is 0.246 mm, etc. Also, L21=d.sub.2 and L51=d.sub.5. The lens elements in Tables 1 and 2 (as well as in Tables 3-6) are designed to provide an image on

an entire 1/3" sensor having dimensions of approximately 4.7×3.52 mm. The optical diameter in all of these lens assemblies is the diameter of the second surface of the fifth lens element.

(34) TABLE-US-00001 TABLE 1 Radius R Distances Diameter # Comment [mm] [mm] Nd/Vd [mm] 1 Stop Infinite -0.466 2.4 2 L11 1.5800 0.894 1.5345/57.095 2.5 3 L12 -11.2003 0.020 2.4 4 L21 33.8670 0.246 1.63549/23.91 2.2 5 L22 3.2281 0.449 1.9 6 L31 -12.2843 0.290 1.5345/57.095 1.9 7 L32 7.7138 2.020 1.8 8 L41 -2.3755 0.597 1.63549/23.91 3.3 9 L42 -1.8801 0.068 3.6 10 L51 -1.8100 0.293 1.5345/57.095 3.9 11 L52 -5.2768 0.617 4.3 12 Window Infinite 0.210 1.5168/64.17 3.0 13 Infinite 0.200 3.0

(35) TABLE-US-00002 TABLE 2 Conic # coefficient k $\alpha_{sub.2}$ $\alpha_{sub.3}$ $\alpha_{sub.4}$ $\alpha_{sub.5}$ $\alpha_{sub.6}$ 2 -0.4668 7.9218E-03 2.3146E-02 -3.3436E-02 2.3650E-02 -9.2437E-03 3 -9.8525 2.0102E-02 2.0647E-04 7.4394E-03 -1.7529E-02 4.5206E-03 4 10.7569 -1.9248E-03 8.6003E-02 1.1676E-02 -4.0607E-02 1.3545E-02 5 1.4395 5.1029E-03 2.4578E-01 -1.7734E-01 2.9848E-01 -1.3320E-01 6 0.0000 2.1629E-01 4.0134E-02 1.3615E-02 2.5914E-03 -1.2292E-02 7 -9.8953 2.3297E-01 8.2917E-02 -1.2725E-01 1.5691E-01 -5.9624E-02 8 0.9938 -1.3522E-02 -7.0395E-03 1.4569E-02 -1.5336E-02 4.3707E-03 9 -6.8097 -1.0654E-01 1.2933E-02 2.9548E-04 -1.8317E-03 5.0111E-04 10 -7.3161 -1.8636E-01 8.3105E-02 -1.8632E-02 2.4012E-03 -1.2816E-04 11 0.0000 -1.1927E-01 7.0245E-02 -2.0735E-02 2.6418E-03 -1.1576E-04

(36) Lens unit **100** provides a field of view (FOV) of 44 degrees, with EFL=6.90 mm, F # =2.80 and TTL of 5.904 mm. Thus and advantageously, the ratio TTL/EFL=0.855. Advantageously, the Abbe number of the first, third and fifth lens element is 57.095. Advantageously, the first air gap between lens elements **102** and **104** (the gap between surfaces **102b** and **104a**) has a thickness (0.020 mm) which is less than a tenth of thickness d.sub.2 (0.246 mm). Advantageously, the Abbe number of the second and fourth lens elements is 23.91. Advantageously, an effective third air gap G (see below with reference to Table 9) between lens elements **106** and **108** (i.e. the telephoto and field lens assemblies) is greater than TTL/5. Advantageously, an effective fourth air gap (see below with reference to Table 9) between lens elements **108** and **110** is smaller than TTL/50.

(37) The focal length (in mm) of each lens element in lens unit **100** is as follows: f1=2.645, f2=-5.578, f3=-8.784, f4=9.550 and f5=-5.290. The condition $1.2 \times |f3| > |f2| > 1.5 \times f1$ is clearly satisfied, as $1.2 \times 8.787 > 5.578 > 1.5 \times 2.645$. f1 also fulfills the condition $f1 < TTL/2$, as $2.645 < 2.952$.

(38) FIG. 3A shows a schematic illustration of an optical lens unit **200**, according to another example of the presently disclosed subject matter. FIG. 3B shows the MTF vs. focus shift of the entire optical lens system for various fields in embodiment **200**. FIG. 3C shows the distortion +Y in percent vs. field.

(39) According to the example illustrated in FIG. 3A, lens unit **200** comprises, in order from an object side to an image side: an optional stop **201**; a telephoto lens assembly including a first plastic lens element **202** with positive refractive power having a convex object-side surface **202a** and a convex or concave image-side surface **202b**, a second plastic lens element **204** with negative refractive power, having a meniscus convex object-side surface **204a**, with an image side surface marked **204b**, and a third plastic lens element **206** with negative refractive power having a concave object-side surface **206a** with an inflection point and a concave image-side surface **206b**; and a field lens assembly including a fourth plastic lens element **208** with positive refractive power having a positive meniscus, with a concave object-side surface marked **208a** and an image-side surface marked **208b**, and a fifth plastic lens element **210** with negative refractive power having a negative meniscus, with a concave object-side surface marked **110a** and an image-side surface marked **210b**. The optical lens unit **200** further optionally includes a glass window **212** disposed between the image-side surface **210b** of fifth lens element **210** and an image plane **214**.

(40) In the lens unit **200**, all lens element surfaces are aspheric. Detailed optical data is given in Table 3, and the aspheric surface data is given in Table 4, wherein the markings and units are the same as in, respectively, Tables 1 and 2. The equation of the aspheric surface profiles is the same as

for lens unit **100** described above.

(41) TABLE-US-00003 TABLE 3 Radius R Distances Diameter # Comment [mm] [mm] Nd/Vd [mm] 1 Stop Infinite -0.592 2.5 2 L11 1.5457 0.898 1.53463/56.18 2.6 3 L12 -127.7249 0.129 2.6 4 L21 6.6065 0.251 1.91266/20.65 2.1 5 L22 2.8090 0.443 1.8 6 L31 9.6183 0.293 1.53463/56.18 1.8 7 L32 3.4694 1.766 1.7 8 L41 -2.6432 0.696 1.632445/23.35 3.2 9 L42 -1.8663 0.106 3.6 10 L51 -1.4933 0.330 1.53463/56.18 3.9 11 L52 -4.1588 0.649 4.3 12 Window Infinite 0.210 1.5168/64.17 5.4 13 Infinite 0.130 5.5

(42) TABLE-US-00004 TABLE 4 Conic # coefficient k $\alpha_{sub.2}$ $\alpha_{sub.3}$ $\alpha_{sub.4}$ $\alpha_{sub.5}$ $\alpha_{sub.6}$ 2 0.0000 -2.7367E-03 2.8779E-04 -4.3661E-03 3.0069E-03 -1.2282E-03 3 -10.0119 4.0790E-02 -1.8379E-02 2.2562E-02 -1.7706E-02 4.9640E-03 4 10.0220 4.6151E-02 5.8320E-02 -2.0919E-02 -1.2846E-02 8.8283E-03 5 7.2902 3.6028E-02 1.1436E-01 -1.9022E-02 4.7992E-03 -3.4079E-03 6 0.0000 1.6639E-01 5.6754E-02 -1.2238E-02 -1.8648E-02 1.9292E-02 7 8.1261 1.5353E-01 8.1427E-02 -1.5773E-01 1.5303E-01 -4.6064E-02 8 0.0000 -3.2628E-02 1.9535E-02 -1.6716E-02 -2.0132E-03 2.0112E-03 9 0.0000 1.5173E-02 -1.2252E-02 3.3611E-03 -2.5303E-03 8.4038E-04 10 -4.7688 -1.4736E-01 7.6335E-02 -2.5539E-02 5.5897E-03 -5.0290E-04 11 0.00E+00 -8.3741E-02 4.2660E-02 -8.4866E-03 1.2183E-04 7.2785E-05

(43) Lens unit **200** provides a FOV of 43.48 degrees, with EFL=7 mm, F # = 2.86 and TTL=5.90 mm. Thus, advantageously, the ratio TTL/EFL=0.843. Advantageously, the Abbe number of the first, third and fifth lens elements is 56.18. The first air gap between lens elements **202** and **204** has a thickness (0.129 mm) which is about half the thickness $d_{sub.2}$ (0.251 mm). Advantageously, the Abbe number of the second lens element is 20.65 and of the fourth lens element is 23.35. Advantageously, the effective third air gap G between lens elements **206** and **208** is greater than TTL/5. Advantageously, the effective fourth air gap between lens elements **208** and **210** is smaller than TTL/50.

(44) The focal length (in mm) of each lens element in lens unit **200** is as follows: $f_1=2.851$, $f_2=-5.468$, $f_3=-10.279$, $f_4=7.368$ and $f_5=-4.536$. The condition $1.2 \times |f_3| > |f_2| > 1.5 \times f_1$ is clearly satisfied, as $1.2 \times 10.279 > 5.468 > 1.5 \times 2.851$. f_1 also fulfills the condition $f_1 < TTL/2$, as $2.851 < 2.950$.

(45) FIG. 4A shows a schematic illustration of an optical lens unit **300**, according to yet a further example of the presently disclosed subject matter. FIG. 4B shows the MTF vs. focus shift of the entire optical lens system for various fields in embodiment **300**. FIG. 4C shows the distortion +Y in percent vs. field.

(46) Lens unit **300** comprises, in order from an object side to an image side, an optional stop **301**; a telephoto lens assembly including a first plastic lens element **302** with positive refractive power having a convex object-side surface **302a** and a convex or concave image-side surface **302b**, a second plastic lens element **204** with negative refractive power, having a meniscus convex object-side surface **304a**, with an image side surface marked **304b**, a third plastic lens element **306** with negative refractive power having a concave object-side surface **306a** with an inflection point and a concave image-side surface **306b**; and a field lens assembly including a fourth plastic lens element **308** with positive refractive power having a positive meniscus, with a concave object-side surface marked **308a** and an image-side surface marked **308b**, and a fifth plastic lens element **310** with negative refractive power having a negative meniscus, with a concave object-side surface marked **310a** and an image-side surface marked **310b**. Also, an optional glass window **312** may be disposed between the image-side surface **310b** of fifth lens element **310** and an image plane **314**.

(47) According to the present example of lens unit **300**, all lens element surfaces are aspheric. Detailed optical data is given in Table 5, and the aspheric surface data is given in Table 6, wherein the markings and units are the same as in, respectively, Tables 1 and 2. The equation of the aspheric surface profiles is the same as for lens units **100** and **200**.

(48) TABLE-US-00005 TABLE 5 Radius R Distances Diameter # Comment [mm] [mm] Nd/Vd [mm] 1 Stop Infinite -0.38 2.4 2 L11 1.5127 0.919 1.5148/63.1 2.5 3 L12 -13.3831 0.029 2.3

4 L21 8.4411 0.254 1.63549/23.91 2.1 5 L22 2.6181 0.426 1.8 6 L31 -17.9618 0.265
 1.5345/57.09 1.8 7 L32 4.5841 1.998 1.7 8 L41 -2.8827 0.514 1.63549/23.91 3.4 9 L42
 -1.9771 0.121 3.7 10 L51 -1.8665 0.431 1.5345/57.09 4.0 11 L52 -6.3670 0.538 4.4 12 Window
 Infinite 0.210 1.5168/64.17 3.0 13 Infinite 0.200 3.0

(49) TABLE-US-00006 TABLE 6 Conic # coefficient k $\alpha_{sub.2}$ $\alpha_{sub.3}$ $\alpha_{sub.4}$ $\alpha_{sub.5}$ $\alpha_{sub.6}$ 2
 -0.534 1.3253E-02 2.3699E-02 -2.8501E-02 1.7853E-02 -4.0314E-03 3 -13.473 3.0077E-02
 4.7972E-03 1.4475E-02 -1.8490E-02 4.3565E-03 4 -10.132 7.0372E-04 1.1328E-01
 1.2346E-03 -4.2655E-02 8.8625E-03 5 5.180 -1.9210E-03 2.3799E-01 -8.8055E-02
 2.1447E-01 -1.2702E-01 6 0.000 2.6780E-01 1.8129E-02 -1.7323E-02 3.7372E-02
 -2.1356E-02 7 10.037 2.7660E-01 -1.0291E-02 -6.0955E-02 7.5235E-02 -1.6521E-02 8
 1.703 2.6462E-02 -1.2633E-02 -4.7724E-04 -3.2762E-03 1.6551E-03 9 -1.456
 5.7704E-03 -1.8826E-02 5.1593E-03 -2.9999E-03 8.0685E-04 10 -6.511 -2.1699E-01
 1.3692E-01 -4.2629E-02 6.8371E-03 -4.1415E-04 11 0.000 -1.5120E-01 8.6614E-02
 -2.3324E-02 2.7361E-03 -1.1236E-04

(50) Lens unit **300** provides a FOV of 44 degrees, EFL=6.84 mm, F # =2.80 and TTL=5.904 mm. Thus, advantageously, the ratio TTL/EFL=0.863. Advantageously, the Abbe number of the first lens element is 63.1, and of the third and fifth lens elements is 57.09. The first air gap between lens elements **302** and **304** has a thickness (0.029 mm) which is about 1/10^{sup.th} the thickness d_{sub.2} (0.254 mm). Advantageously, the Abbe number of the second and fourth lens elements is 23.91. Advantageously, the effective third air gap G between lens elements **306** and **308** is greater than TTL/5. Advantageously, the effective fourth air gap between lens elements **308** and **310** is smaller than TTL/50.

(51) The focal length (in mm) of each lens element in embodiment **300** is as follows: f₁=2.687, f₂=-6.016, f₃=-6.777, f₄=8.026 and f₅=-5.090. The condition 1.2×|f₃|>|f₂|>1.5×f₁ is clearly satisfied, as 1.2×6.777>6.016>1.5×2.687. f₁ also fulfills the condition f₁<TTL/2, as 2.687<2.952.

(52) Tables 7 and 8 provide respectively detailed optical data and aspheric surface data for a fourth embodiment of an optical lens system disclosed herein. The markings and units are the same as in, respectively, Tables 1 and 2. The equation of the aspheric surface profiles is the same as for lens systems **100**, **200** and **300**. The lens elements in Tables 7 and 8 are designed to provide an image on an entire ¼" sensor having dimensions of approximately 3.66×2.75 mm.

(53) TABLE-US-00007 TABLE 7 Radius R Distances Diameter # Comment [mm] [mm] Nd/Vd
 [mm] 1 Stop Infinite -0.427 2.1 2 L11 1.3860 0.847 1.534809/55.66 2.2 3 L12 -8.5270 0.073
 2.1 4 L21 11.1443 0.239 1.639078/23.253 1.9 5 L22 1.8641 0.504 1.7 6 L31 19.7342 0.239
 1.534809/55.66 1.7 7 L32 3.9787 1.298 1.7 8 L41 -3.3312 0.522 1.639078/23.253 2.8 9 L42
 -1.7156 0.079 3.1 10 L51 -1.7788 0.298 1.534809/55.66 3.5 11 L52 -12.6104 0.792 3.7 12
 Window Infinite 0.210 1.5168/64.17 4.5 13 Infinite 0.177 4.6

(54) TABLE-US-00008 TABLE 8 Conic # coefficient k α_2 α_3 α_4 α_5 α_6 2 -0.326 8.776E-03
 2.987E-02 -6.001E-02 6.700E-02 -2.849E-02 3 -10.358 4.266E-02 -2.240E-02
 2.914E-02 -3.025E-02 3.108E-03 4 11.447 -3.257E-02 9.780E-02 -1.143E-02
 -3.844E-02 1.005E-02 5 -0.026 -3.631E-02 2.928E-01 -2.338E-01 3.334E-01 -2.760E-02
 6 0.000 1.578E-01 -2.229E-02 -4.991E-02 1.663E-01 -1.298E-01 7 3.860 2.044E-01
 5.451E-02 -3.199E-01 5.619E-01 -3.663E-01 8 4.094 3.706E-02 -5.931E-02 4.662E-02
 -4.654E-02 1.606E-02 9 -9.119 -7.980E-02 -1.376E-03 5.622E-03 -6.715E-03
 2.127E-03 10 -12.777 -2.695E-01 1.894E-01 -5.690E-02 8.689E-03 -5.269E-04 11 0.000
 -1.807E-01 1.278E-01 -4.504E-02 6.593E-03 -2.357E-04

(55) The focal length (in mm) of each lens element according to this example is as follows: f₁=2.298, f₂=-3.503, f₃=-9.368, f₄=4.846 and f₅=-3.910. The condition 1.2×|f₃|>|f₂|>1.5×f₁ is clearly satisfied, as 1.2×9.368>3.503>1.5×2.298. f₁ also fulfills the condition f₁<TTL/2, as 2.298<2.64.

(56) Generally, with regard to the effective air gap between the adjacent lens elements, the

following should be noted.

(57) In each one of the lens units exemplified above, the first three lens elements (L1, L2 and L3) achieve essentially a telephoto effect for all fields (angles of object orientation relative to the optical axis), i.e. achieve a strong concentration (by L1) followed by partial collimation (mainly by L2 but also by L3). The fact that all fields need to have essentially the same telephoto effect leads to relatively small distances (small air gaps) between the three lens elements, e.g. especially between L1 and L2 (air gap 1). L4 and L5 are mainly field lens elements for reducing field curvature, i.e. their main effect is to cause the focal point for all fields (where the object distance is approximately infinity) to reside on the sensor plane. To achieve this, it is advantageous that for every field, the corresponding rays hit L4 and L5 at different locations, thus enabling separate adjustment for every field (“field separation”).

(58) The inventors have found that the desired fields' separation is obtainable in a lens unit design characterized by an “effective air gap” G between lenses L3 and L4 (between the telephoto and field lens assemblies, where a larger G leads to larger separation between the fields).

(59) FIG. 5 illustrates the concept of the effective air gap between the two adjacent lens elements. First, an “air gap per field” $D_{\text{sub.f-n}}$ is defined as the length of the n.sup.th field's chief ray along the respective chief ray between adjacent lens elements. Effective gap $D_{\text{sub.Leff}}$ is then defined as the average of N air gaps per field for field angles α separated evenly between $\alpha=0$ (for ray 1, air gap $D_{\text{sub.f-1}}$) to $\alpha=\alpha_{\text{sub.max}}$ (for ray N, air gap $D_{\text{sub.f-n}}$), where ray N hits the end pixel on the image sensor diagonal. In other words, between each pair of adjacent lens elements (e.g. between L3 and L4 and between L4 and L5):

$$(60) D_{\text{Leff}} = (\text{Math}.\sum_{n=1}^N D_{f-n}) / N$$

(61) In essence, the effective air gap between adjacent lens elements reflects an average effective distance between the two surfaces bounding the air gap between the two adjacent lens elements. Exemplarily, in FIG. 5 there are N=9 chief rays (and 9 related field air gaps) and the chief rays are distributed angularly evenly between $\alpha=0$ for ray 1 and $\alpha_{\text{sub.max}}$ for ray 9. At $\alpha_{\text{sub.max}}$, ray 9 hits the end pixel on the image sensor diagonal.

(62) Table 9 shows data on TTL, $D_{\text{sub.Leff-3}}$, $D_{\text{sub.Leff-4}}$, and ratios between the TTL and the effective air gaps for each of lens units **100**, **200** and **300** above. $D_{\text{sub.Leff-3}}$ and $D_{\text{sub.Leff-4}}$ were calculated using 9 chief rays, as shown in FIG. 4.

(63) TABLE-US-00009 TABLE 9 Embodi- ment TTL $D_{\text{sub.Leff-3}}$ = G $D_{\text{sub.Leff-4}}$ $D_{\text{sub.Leff-3}}/\text{TTL}$ $D_{\text{sub.Leff-4}}/\text{TTL}$

100	5.903	1.880	0.086	0.319	0.015
200	5.901	1.719	0.071	0.291	0.012
300	5.904	1.925	0.094	0.326	0.016
400	5.279	1.263	0.080	0.246	0.015

Using $D_{\text{sub.Leff-3}}=G$ instead of the commonly used distance along the optical axis between L3 and L4 ensures better operation (for the purpose of reduction of field curvature) of lens elements L4 and L5 for all the fields. As seen in Table 9, good field separation may exemplarily be achieved if $D_{\text{sub.Leff-3}}=G>\text{TTL}/5$.

(64) A compact optical design requires that the diameter of L5 be as small as possible while providing the required performance. Since the lens and camera footprint is determined by L5 diameter, a small effective air gap, $D_{\text{sub.Leff-4}}$, between lenses L4 and L5 is advantageous in that it allows a small diameter of lens L5 without degrading the optical performance. Effective air gap $D_{\text{sub.Leff-4}}$ is a better indicator of the L5 diameter than the commonly used air gap along the optical axis between L4 and L5. An adequately small L5 diameter may exemplarily be achieved if the effective air gap between the field lenses L4 and L5 is $D_{\text{sub.Leff-4}}<\text{TTL}/50$. It should be noted that an effective air gap $D_{\text{sub.Leff}}$ can be calculated in principle using any combination of two or more chief rays (for example ray 1 and ray 9 in FIG. 4). However, the “quality” of $D_{\text{sub.Leff}}$ calculation improves while considering an increased number of chief rays.

(65) The miniature telephoto lens units described above with reference to FIGS. 1C and 2 to 5 are designed with a TTL shorter than EFL. Accordingly, due to shorter TTL, such lens units have a smaller field of view, as compared to standard mobile phone lens units. Therefore, it would be

particularly useful to use such a telephoto lens unit as a Tele sub-camera lens unit in a dual aperture zoom camera. Such a dual aperture zoom camera is described in the above-mentioned WO14199338 of the same assignee as the present application.

(66) As mentioned above, a problem associated with the use of conventional Wide and Tele lens modules in a camera is associated with the different lengths/heights of the lenses which can cause shadowing and light blocking effects. According to the presently disclosed subject matter it is suggested to eliminate or at least significantly reduce these shadowing and light blocking effects by replacing the conventional Tele lens module by the miniature telephoto lens unit described above in the dual aperture camera.

(67) Thus, according to the presently disclosed subject matter, the problem discussed above posed by a difference in the TTL/EFL ratios of the conventional Tele and Wide lenses may be solved through use of a standard lens for the Wide camera ($TTL.sub.W/EFL.sub.W > 1.1$, typically 1.3) and of a special Telephoto lens design for the Tele camera ($TTL.sub.T/EFL.sub.T < 1$, e.g. 0.87), where the telephoto lens unit is configured as described above, providing the miniature telephoto lens unit.

(68) Using the above described miniature telephoto lens unit enables to reduce the $TTL.sub.T$ (according to one non-limiting example down to $7 \times 0.87 = 6.09$ mm) leading to a camera height of less than 7 mm (which is an acceptable height for a smartphone or any other mobile electronic device). The height difference between the telephoto lens unit and the Wide lens unit is also reduced to approximately 1.65 mm, thus reducing shadowing and light blocking problems.

(69) According to some examples of a dual-aperture camera disclosed herein, the ratio “e” = $EFL.sub.T/EFL.sub.W$ is in the range 1.3-2.0. In some embodiments, the ratio $TTL.sub.T/TTL.sub.W < 0.8e$. In some embodiments, $TTL.sub.T/TTL.sub.W$ is in the range 1.0-1.25. According to some examples disclosed herein, $EFL.sub.W$ may be in the range 2.5-6 mm and $EFL.sub.T$ may be in the range 5-12 mm.

(70) Referring now to the figures, FIG. 6A shows schematically in perspective cross section an example of a dual-aperture zoom camera device **600**. Camera device **600** includes two camera units **602** and **604**. It should be understood that the two camera units may be associated with common or separate detectors (pixel matrix and their associated read out circuits). Thus, the two camera units are actually different in their optics, i.e. in the imaging channels defined by the wide and telephoto lens units. Each camera unit may be mounted on a separate PCB (respectively **605a** and **605b**) including the read out circuit, and includes a lens unit (respectively **606** and **608**), and an image sensor including a pixel matrix (respectively **614** and **616**), and an actuator (respectively **610** and **612**) associated with a focusing mechanism. In this embodiment, the two PCBs lie in the same plane. It should be understood that in the embodiment where the readout circuits of the two imaging channels are in the same plane, a common PCB can be used, as will be described further below. The two camera units are connected by a case **618**. For example, camera **602** includes a Wide lens unit and camera **604** includes a Telephoto lens unit, the $TTL.sub.T$ of the lens unit defining the respective camera height H. For example, the Wide and Telephoto lens units provide respectively main and auxiliary optical/imaging paths, enabling to use the main image for interpreting the auxiliary image data.

(71) FIG. 6B shows schematically, in perspective cross, another example of a dual-aperture zoom camera **600'** utilizing the principles of the invention. Camera **600'** is generally similar to the above-described camera **600**, and the common components are shown in the figure in a self-explanatory manner and thus are not indicated by reference numbers. As in camera **600**, in the camera **600'**, the camera unit **602** (e.g. a Wide lens camera) and camera unit **604** (e.g. a Telephoto lens camera) are mounted on separate PCBs (respectively **605a** and **605b**). However, in contrast with camera **600**, in camera **600'** the two PCBs lie in different planes. This enables the object side principal planes of the Wide and Telephoto lens units to be close one to the other, thus reducing the dependency of magnification factor in the two units on the object distance.

(72) For example, camera dimensions for the cameras shown in FIGS. 6A and 6B may be as follows: a length L of the camera (in the Y direction) may vary between 13-25 mm, a width W of the camera (in the X direction) may vary between 6-12 mm, and a height H of the camera (in the Z direction, perpendicular to the X-Y plane) may vary between 4-12 mm. More specifically, considering a smartphone camera example disclosed herein, L=18 mm, W=8.5 mm and H=7 mm.

(73) FIG. 7 shows schematically, in perspective cross section, yet another example of a dual-aperture zoom camera **700**. Camera **700** is similar to cameras **600** and **600'** in that it includes two camera units **702** and **704** with respective lens units **706** and **708**, respective actuators **710** and **712** and respective image sensors **714** and **716**. However, in camera **700**, the two camera units **702** and **704** are mounted on a single (common) PCB **705**. The mounting on a single PCB and the minimizing of a distance “d” between the two camera units minimizes and may even completely avoid camera movement (e.g. associated with mishaps such as drop impact). In general, the dimensions of camera **700** may be in the same range as those of cameras **600** and **600'**. However, for the same sensors and optics, the footprint W×L and the weight of camera **700** are smaller than that of cameras **600** and **600'**. Mishaps such as drop impact may cause a relative movement between the two cameras after system calibration, changing the pixel matching between the Tele and Wide images and thus preventing fast reliable fusion of the Tele and Wide images. Therefore, such effects should preferably be eliminated.

(74) As described above, the high-quality imaging is also associated with the implementation of standard optical image stabilization (OIS) in such a dual-aperture zoom camera. Standard OIS compensates for camera tilt (“CT”), i.e., image blur, by a parallel-to-the image sensor (exemplarily in the X-Y plane) lens movement (“LMV”). The amount of LMV (in millimeters) needed to counter a given camera tilt depends on the camera lens EFL, according to the relation:

$$\text{LMV} = \text{CT} * \text{EFL},$$

where “CT” is in radians and EFL is in mm.

(75) Since the Wide and telephoto lens units have significantly different EFLs, both lenses cannot move together and achieve optimal tilt compensation for both of the respective camera units. More specifically, since the tilt is the same for both camera units, a movement that will compensate for the tilt for the Wide camera unit will be insufficient to compensate for the tilt for the Telephoto camera unit, and vice versa. Using separate OIS actuators for the two camera units respectively can achieve simultaneous tilt compensation for both of them, but the entire system would be complex and costly, which is undesirable for portable electronic devices.

(76) In this connection, reference is made to FIG. 8 which shows an example of a dual-aperture zoom camera **800** (similar to the above-described camera **700**) that includes two camera units **802** and **804** mounted either on a single PCB **805** (as shown in this example) or on separate PCBs. Each camera unit includes a lens unit (respectively **806** and **808**), an actuator (respectively **810** and **812**) and an image sensor (respectively **814** and **816**). The two actuators are rigidly mounted on a rigid base **818** that is flexibly connected to the PCB (or PCBs) through flexible elements **820**. Base **818** is movable by an OIS mechanism (not shown) controlled by an OIS controller **902** (shown in FIG. 9). The OIS controller **902** is coupled to, and receives camera tilt information from a tilt sensor (e.g. a gyroscope) **904** (FIG. 9). More details of an example of an OIS procedure as disclosed herein are given below with reference to FIG. 9. The two camera units are separated by a small distance “d”, e.g. 1 mm. This small distance between camera units also reduces the perspective effect enabling smoother zoom transition between the camera units.

(77) As indicated above, the two image sensors **814** and **816** may be mounted on separate PCBs that are rigidly connected, thereby enabling adaptation of an OIS mechanism to other system configurations, for example those described above with reference to FIGS. 6A and 6B.

(78) In some embodiments, and optionally, a magnetic shield plate may be used, e.g. as described in co-owned U.S. patent application Ser. No. 14/365,718 titled “Magnetic shielding between voice coil motors in a dual-aperture camera”, which is incorporated herein by reference in its entirety.

Such a magnetic shield plate may be inserted in the gap (with width d) between the Wide and Tele camera units.

(79) In general, the dimensions of camera **800** may be in the same range as those of cameras **600**, **600'** and **700**.

(80) Reference is made to FIG. **9**, which exemplifies the camera operation, utilizing a tilt sensor **904** which dynamically measures the camera tilt (which is the same for both the Wide and Tele camera units). As shown, an OIS controller **902** (electronic circuit including hardware/software components) is provided, which is coupled to the actuators of both camera units (e.g. through base **818**), and receives a CT input from the tilt sensor **904** and a user-defined zoom factor, and controls the lens movement of the two camera units to compensate for the tilt. The LMV is for example in the X-Y plane. The OIS controller **902** is configured to provide a LMV equal to $CT * EFL_{sub.ZF}$, where "EFL.sub.ZF" is chosen according to the user-defined zoom factor, ZF. According to one example of an OIS procedure, when $ZF=1$, LMV is determined by the Wide camera unit's EFL.sub.W (i.e. $EFL_{sub.ZF}=EFL_{sub.W}$ and $LMV=CT * EFL_{sub.W}$). Further, when $ZF>e$ (i.e. $ZF>EFL_{sub.T}/EFL_{sub.W}$), LMV is determined by the telephoto camera unit's EFL.sub.T (i.e. $EFL_{sub.ZF}=EFL_{sub.T}$ and $LMV=CT * EFL_{sub.T}$). Further yet, for a ZF between 1 and e, the EFL.sub.ZF may shift gradually from EFL.sub.W to EFL.sub.T according to $EFL_{sub.ZF}=ZF * EFL_{sub.W}$.

(81) Thus, the present invention provides a novel approach for configuring a camera device suitable for use in portable electronic devices, in particular smart phones. The present invention solves various problems associated with the requirements for physical parameters of such devices (weight, size), high image quality and zooming effects.

Claims

1. A dual-aperture camera, comprising: a Wide camera having a Wide lens with a field of view $FOV_{sub.W}$ and having a total track length $TTL_{sub.W}$ and an effective focal length $EFL_{sub.W}$, and a Telephoto camera comprising a Telephoto lens with a Telephoto field of view $FOV_{sub.T}$, a total track length $TTL_{sub.T}$ and an effective focal length $EFL_{sub.T}$, wherein the $TTL_{sub.T}$ includes a glass window, wherein a ratio $TTL_{sub.W}/EFL_{sub.W}$ is greater than 1, wherein a ratio $TTL_{sub.T}/EFL_{sub.T}$ is smaller than 1, wherein the Telephoto lens has an optical axis and a F # smaller than 3.2 and includes, in order from an object plane to an image plane along the optical axis, a first lens element having positive optical power and two last lens elements, wherein one of the two last lens elements has an Abbe number greater than 50, wherein the other of the two last lens elements has an Abbe number smaller than 30, and wherein a height H of the dual-aperture camera measured along the optical axis is in the range of 4 mm-12 mm.
2. The dual-aperture camera of claim 1, wherein $EFL_{sub.T}$ is in the range $EFL_{sub.T}=5-12$ mm.
3. The dual-aperture camera of claim 1, wherein F # is smaller than 2.9.
4. The dual-aperture camera of claim 1, wherein $TTL_{sub.T}/EFL_{sub.T}$ is smaller than 0.9.
5. The dual-aperture camera of claim 1, wherein the Telephoto lens unit includes a pair of lens elements formed by a second lens element and a third lens element, and wherein the pair of lens elements has a combined negative optical power.
6. The dual-aperture camera of claim 1, wherein the first lens element of the Telephoto lens unit has a focal length f_1 smaller than $TTL_{sub.T}/2$.
7. The dual-aperture camera of claim 1, wherein the two last lens elements are spaced apart from one another by an effective air gap smaller than $TTL_{sub.T}/50$.
8. The dual-aperture camera of claim 1, wherein the first, second and third lens elements have respective focal lengths f_1 , f_2 and f_3 , and wherein the respective focal lengths satisfy the condition $1.2|f_3|>|f_2|>1.5f_1$.
9. The dual-aperture camera of claim 1, wherein the Telephoto camera unit includes an image

sensor with an image sensor size $\frac{1}{4}$ " to $\frac{1}{3}$ ".

10. The dual-aperture camera of claim 1, wherein the Wide and Telephoto camera units are spaced from one another a distance of about 1 mm.

11. The dual-aperture camera of claim 1, wherein TTL.sub.T is smaller than 5.5 mm.

12. The dual-aperture of claim 1, included in a mobile electronic device.

13. The dual-aperture camera of claim 12, wherein the mobile device is a smartphone.

14. The dual-aperture of claim 2, included in a mobile electronic device.

15. The dual-aperture camera of claim 14, wherein the mobile device is a smartphone.

16. The dual-aperture of claim 3, included in a mobile electronic device.

17. The dual-aperture camera of claim 16, wherein the mobile device is a smartphone.

18. The dual-aperture camera of claim 4, included in a mobile electronic device.

19. The dual-aperture camera of claim 18, wherein the mobile device is a smartphone.
